Stat 220: Probability

Midterm practice questions, with answers

Practice on the three rules, conditioning, reversing a conditional, expected value and variance, and the behaviour of averages. Roughly half of these want a number, and the arithmetic is small enough to do in the margin. No computer needed.

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1. A batch job fails on any given night with probability 0.02, and nights are independent. Over a 60-night quarter, what is the chance of at least one failure?

   A. $0.02 \times 60 = 1.20$, so a failure is certain.

   B. $1 - 0.02^{60}$, which is essentially 1.

   C. $0.98^{60} = 0.30$.

   D. $1 - 0.98^{60} = 0.70$.

   Answer: D. Anything phrased "at least one" is easier backwards. $\mathbb{P}(\text{none}) = 0.98^{60} = 0.296$, so at least one has probability 0.704. Note what multiplying the nightly rate by 60 would have given: probabilities do not add across trials, and 1.20 is not a probability at all. A 2% nightly risk is worse than even odds over a quarter.

2. $\mathbb{P}(A) = 0.40$, $\mathbb{P}(B) = 0.25$, and $\mathbb{P}(A \text{ and } B) = 0.10$. What is $\mathbb{P}(A \text{ or } B)$?

   A. 0.65

   B. 0.75

   C. 0.10

   D. 0.55

   Answer: D. $0.40 + 0.25 - 0.10 = 0.55$. The overlap gets counted once in each of the first two terms, so it has to come back out once. Adding to 0.65 is the error of treating the two as mutually exclusive, which the positive joint already rules out.

3. Ten servers are in a rack and four of them have been patched. Two are picked at random, one after the other, without replacement. What is the chance both are patched?

   A. $(4/10)^2 = 0.16$

   B. $\frac{4}{10} \times \frac{3}{9} = 0.133$

   C. $4/10 + 3/9 = 0.73$

   D. $4/10 = 0.40$, since the second pick does not change the proportion.

   Answer: B. Use the multiplication rule with the condition doing real work: the first is patched with probability 4/10, and given that, only 3 of the remaining 9 are patched. Squaring 4/10 is

the independence shortcut, and here independence is false by construction, because removing a server changes the pool.

4. Which of these statements is true for *every* pair of events, with no extra assumption?

   A. $\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\,\mathbb{P}(B)$

   B. $\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\,\mathbb{P}(B \mid A)$

   C. $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)$

   D. $\mathbb{P}(A \text{ or } B) = \mathbb{P}(A) + \mathbb{P}(B)$

   Answer: B. The multiplication rule in its conditional form is always true, and it is the definition of conditional probability rearranged. Multiplying the two margins needs independence; adding them without subtracting the overlap needs the events to be mutually exclusive; and the two conditionals being equal is almost never true.

5. Of 1,000 support tickets, 300 came from enterprise accounts and 700 from everyone else. 120 of the enterprise tickets were escalated, and 70 of the others were. What is $\mathbb{P}(\text{escalated} \mid \text{enterprise})$?

   A. $120/1000 = 0.12$

   B. $120/190 = 0.63$

   C. $190/1000 = 0.19$

   D. $120/300 = 0.40$

   Answer: D. Conditioning on enterprise throws away the other 700 rows, so the denominator is 300, not 1,000 and not the total escalated. Dividing by 1,000 gives the joint probability $\mathbb{P}(\text{escalated and enterprise}) = 0.12$, and dividing by 190 answers the reversed question, $\mathbb{P}(\text{enterprise} \mid \text{escalated})$.

6. Same 1,000 tickets. What is $\mathbb{P}(\text{enterprise} \mid \text{escalated})$?

   A. 0.40

   B. 0.12

   C. 0.63

   D. 0.19

   Answer: C. 190 tickets were escalated in total, and 120 of them were enterprise, so $120/190 = 0.63$. Same numerator as the previous question, different denominator, different answer. This pair is the whole point of keeping the direction straight.

7. A joint table gives $\mathbb{P}(A) = 0.30$, $\mathbb{P}(B) = 0.50$, and $\mathbb{P}(A \text{ and } B) = 0.21$. Are A and B independent?

A. No. Independence needs the joint to be $0.30 \times 0.50 = 0.15$, and it is 0.21.

B. Yes, because 0.21 is close to 0.30 and 0.50 is close to 0.21 doubled.

C. Yes, because the joint is positive and neither margin is zero.

D. There is not enough information; you also need the sample size.

Answer: A. Independence is an arithmetic check: compare the joint against the product of the margins. Equivalently $\mathbb{P}(A \mid B) = 0.21/0.50 = 0.42$, which is not $\mathbb{P}(A) = 0.30$. Knowing B happened raises the chance of A by 12 points, so they are dependent.

8. Two events each have probability 0.30 and are mutually exclusive. What is $\mathbb{P}(A \mid B)$?

A. 0, because if B happened then A did not.

B. 0.09

C. 0.30, since mutual exclusivity does not affect conditional probabilities.

D. 0.60

Answer: A. $\mathbb{P}(A \text{ and } B) = 0$, so the conditional is $0/0.30 = 0$. Learning B took A from 0.30 to 0, which is as far as evidence can move anything. Mutually exclusive events are strongly dependent, and 0.09 is the answer you would get by assuming the opposite.

9. Can two events with probabilities 0.30 and 0.50 be both mutually exclusive and independent?

A. Yes, they are different properties and nothing stops both holding.

B. No. Mutually exclusive forces the joint to 0, and independent forces it to 0.15.

C. Yes, but only if the sample space has exactly two outcomes.

D. Only if one of them is the complement of the other.

Answer: B. A joint probability cannot be both 0 and 0.15. Any two events with positive probabilities can be one or the other, never both. The two words get used interchangeably in practice and they are closer to opposites.

10. A condition affects 1% of people. The test catches 95% of those who have it, and gives a false positive for 10% of those who do not. Someone tests positive. What is the chance they have the condition?

A. About 9%.

B. About 50%, since a positive is equally likely either way.

C. About 95%, since that is the accuracy of the test.

D. About 1%, since the test tells you nothing beyond the base rate.

Answer: A. Count 10,000 people. 100 have it, and 95 of those test positive. 9,900 do not, and 990 of those test positive anyway. That is 1,085 positives, of which 95 are real: $95/1085 = 0.088$. The false positives come from a group ninety-nine times larger, which is why they swamp the true ones. The test is still informative, it moved the number from 1% to 9%.

11. In the previous question, which number would do the most to raise the chance that a positive is real?

    A. Raising the 95% detection rate to 99%.

    B. Testing more people.

    C. Cutting the 10% false positive rate to 2%.

    D. Running the same test a second time on everyone.

    Answer: C. The false positives dominate because they are drawn from the 99% who are healthy. At a 2% false positive rate the arithmetic becomes $95/(95+198) = 0.32$, a fourfold improvement. Pushing detection from 95 to 99 moves the numerator by four people and changes almost nothing. When the base rate is low, specificity is the number that matters.

12. "The chance of a match this close if the suspect were innocent is one in a million, so there is a one in a million chance he is innocent." What is wrong with this?

    A. It reports $\mathbb{P}(\text{evidence} \mid \text{innocent})$ and reads it as $\mathbb{P}(\text{innocent} \mid \text{evidence})$.

    B. The one in a million figure is probably too small.

    C. Nothing; the two statements are equivalent.

    D. It should have used the addition rule rather than the multiplication rule.

    Answer: A. This is the prosecutor's fallacy, and it is the Sally Clark case. Reversing the conditional needs Bayes' rule, and that brings in how many people could have produced the evidence. In a pool of thirty million, a one in a million match rate produces about thirty candidates.

13. A support ticket needs X callbacks, where X is 0, 1, 2 or 3 with probabilities 0.50, 0.30, 0.15 and 0.05. What is $\mathbb{E}[X]$?

    A. 0.75

    B. 1.5, the midpoint of the range.

    C. 0.25, since each outcome is a quarter.

    D. 1.0

    Answer: A. $0(0.50) + 1(0.30) + 2(0.15) + 3(0.05) = 0.30 + 0.30 + 0.15 = 0.75$. The expected value weights each outcome by how often it happens, so it sits well below the midpoint when the small values carry most of the probability. No ticket ever needs 0.75 callbacks.

14. Same distribution. What is the variance of X?

    A. 0.75

    B. 0.89

    C. 0.7875

    D. 1.35

    Answer: C. $\mathbb{E}[X^2] = 0(0.50) + 1(0.30) + 4(0.15) + 9(0.05) = 1.35$, so $\mathrm{Var}(X) = 1.35 - 0.75^2 = 1.35 - 0.5625 = 0.7875$. Stopping at 1.35 is the error of forgetting to subtract the squared mean, and 0.89 is the standard deviation, $\sqrt{0.7875}$, which is the one you would report because it is in callbacks.

15. Each callback costs \$18 and there is a \$40 fixed cost per ticket, so the total is $Y = 40 + 18X$ with $\mathbb{E}[X] = 0.75$ and $\mathrm{Var}(X) = 0.7875$. What are $\mathbb{E}[Y]$ and $\mathrm{SD}(Y)$?

    A. \$53.50 and \$14.18

    B. \$53.50 and \$15.97

    C. \$58.00 and \$15.97

    D. \$53.50 and \$0.79

    Answer: B. $\mathbb{E}[Y] = 40 + 18(0.75) = 53.50$ by linearity. Variance ignores the shift and squares the multiplier: $\mathrm{Var}(Y) = 18^2(0.7875) = 255.15$, so the standard deviation is \$15.97. Adding a constant moves the centre and leaves the spread alone.

16. A roadside call costs the company \$280 in standard deviation. Two independent calls come in. What is the standard deviation of the *total* cost of the two?

    A. \$560, since standard deviations add.

    B. \$140

    C. \$280, since the calls are independent.

    D. \$396

    Answer: D. Variances add for independent draws, standard deviations do not. Var $= 280^2 + 280^2 = 156{,}800$, and $\sqrt{156{,}800} = 396$, which is $280\sqrt{2}$. Adding the standard deviations overstates the spread by 41%, and it is the single most common version of this mistake.

17. X is \$1 or \$9, each with probability one half. A contract pays $\sqrt{X}$ thousand dollars. What does it pay on average, and how does that compare to $\sqrt{\mathbb{E}[X]}$?

    A. \$2.24k on average, which equals $\sqrt{5}$.

    B. \$2.00k on average, which is below $\sqrt{5} = 2.24$.

C. \$2.00k on average, which is above $\sqrt{5} = 2.24$.

D. \$5.00k on average, since the mean of X is 5.

Answer: B. $\mathbb{E}[\sqrt{X}] = \frac{1}{2}(1) + \frac{1}{2}(3) = 2$, while $\sqrt{\mathbb{E}[X]} = \sqrt{5} = 2.24$. Putting the average input through a curved function is not the same as averaging the outputs, and for a function that bends downward the plug-in answer is too high. Planning on the average input is how projects get schedules that cannot be met.

18. A hardware warranty claim costs the manufacturer nothing 80% of the time, \$200 15% of the time, and \$4,000 the remaining 5%. What is the expected cost of a claim?

A. \$1,400

B. \$200

C. \$230

D. \$30

Answer: C. $0(0.80) + 200(0.15) + 4000(0.05) = 0 + 30 + 200 = \$230$. Note where it comes from: the \$4,000 outcome lands on one claim in twenty and supplies \$200 of the \$230, 87% of the total. Four claims in five cost nothing at all, so no claim ever costs the expected amount, and the median claim costs \$0. Averaging the three dollar figures without their probabilities gives \$1,400 and throws away how rare the expensive one is.

19. Routes fall into three groups: 50% are short and average 100 minutes, 30% are medium and average 200, and 20% are long and average 400. What is the overall mean?

A. 233, the average of 100, 200 and 400.

B. 190

C. 700

D. 200, the middle group's mean.

Answer: B. $0.50(100) + 0.30(200) + 0.20(400) = 50 + 60 + 80 = 190$. The overall mean is a weighted average of the conditional means, weighted by how often each condition occurs. Averaging the three group means without weights gives 233 and is wrong whenever the groups differ in size.

20. A regression tree splits on `stops` and reports 405 minutes in one leaf. In the language of this unit, what is 405?

A. $\mathbb{E}[Y]$, estimated on the rows in that leaf.

B. $\mathbb{P}(Y \mid X)$ for that leaf.

C. The variance of Y within that leaf.

D. $\mathbb{E}[Y \mid X$ in that leaf], estimated by the average of those rows.

Answer: D. A tree is a conditional expectation with the conditioning set chosen by the algorithm. Each leaf defines a condition, and the number it reports is the mean of the outcome given that condition. Reading a fitted value as a conditional expectation is what connects every model in this course back to this unit.

21. Service calls have $\sigma = \$280$. A company handles 400 of them in a day. What is the standard deviation of the *average* cost per call that day?

    A. \$14

    B. \$0.70

    C. \$280

    D. \$5,600

    Answer: A. $\sigma/\sqrt{n} = 280/20 = 14$. Dividing by n rather than $\sqrt{n}$ gives \$0.70 and is far too optimistic. The \$5,600 figure is the standard deviation of the *total*, $280\sqrt{400}$, which grows while the average's shrinks.

22. A survey of 100 people gives a standard error of 3 points. How many people would be needed to get it down to 1.5 points?

    A. 400

    B. 200

    C. 150

    D. 10,000

    Answer: A. The standard error falls with $\sqrt{n}$, so halving it costs four times the data. Doubling to 200 only gets you to $3/\sqrt{2} = 2.1$. This is why the step from a rough answer to a precise one is so much more expensive than the step from nothing to a rough answer.

23. As n grows, which of these shrinks?

    A. The standard deviation of the individual observations.

    B. Both, at the same rate.

    C. The standard error of the sample mean.

    D. Neither; both are properties of the population.

    Answer: C. σ is a fact about one draw and does not care how many draws you take. The standard error is a fact about the average and falls as $\sigma/\sqrt{n}$. Collecting more data pins down the mean, it does not make individual outcomes more alike, and conflating the two produces prediction intervals that are far too narrow.

24. Someone has $n = 50$ and says the central limit theorem applies because 50 is over 30. What is the problem with that reasoning?

    A. Nothing; 30 is the standard threshold.

    B. The threshold is 100, not 30.

    C. The central limit theorem needs the population to be normal first.

    D. How large n needs to be depends on how skewed the distribution is, and 30 is not a universal number.

    Answer: D. For a mild distribution 10 is plenty; for the service call table, where one outcome in a hundred carries a quarter of the mean, the average of 30 is still visibly skewed. The rule of thumb is a summary of the easy case. Simulate from your own distribution rather than checking n against a number someone memorised.

25. The central limit theorem says the sample mean is approximately normal for large n. Which of these does it *not* tell you?

    A. That the sample mean has standard deviation $\sigma/\sqrt{n}$.

    B. That the sample mean is centred on μ.

    C. That the individual observations are approximately normal.

    D. That the spread of the sample mean shrinks as n grows.

    Answer: C. It is a statement about the average, not about the data. The service calls stay as skewed as they ever were no matter how many you collect. Treating the raw data as normal because n is large is a different claim and usually a false one.